



Session 4: Building Data Models

Objectives

- Explain the concept of data modeling and its importance in Power BI.
- Demonstrate normalization techniques to structure data efficiently.
- Illustrate the creation of star schema data models using the AdventureWorks 2022 database.
- Build relationships between tables to enable integrated analysis.
- Apply data modeling principles to a custom dataset for practical understanding.

Definition of Terms

- **Data Modeling:** The process of structuring and organizing data tables to optimize analysis and reporting in Power BI.
- **Normalization:** A technique to eliminate data redundancy and ensure data integrity by organizing data into related tables with minimal duplication.
- **Star Schema:** A data modeling approach with a central fact table (containing transactional data) connected to dimension tables (containing descriptive data) via relationships.
- **Fact Table:** A table in a star schema that stores quantitative data (e.g., sales amounts, quantities) linked to dimension tables.
- **Dimension Table:** A table in a star schema that stores descriptive attributes (e.g., product names, customer details) related to the fact table.
- **Relationships:** Connections between tables based on common columns (e.g., keys) to enable integrated queries and visualizations.
- **Primary Key:** A unique identifier for each row in a table (e.g., ProductID in a Product table).
- **Foreign Key:** A column in one table that references the primary key of another table to establish a relationship.

Sample Dataset

Below is a custom dataset designed to demonstrate data modeling concepts, including

normalization and star schema design, complementing the AdventureWorks 2022 database.

RawSalesData Table (Pre-Normalization)

This table simulates unnormalized sales data with redundant information.

OrderID	CustomerName	CustomerCity	ProductName	ProductCategory	SaleDate
O001	John Smith	New York	Laptop	Electronics	2023-07-01
O002	John Smith	New York	Mouse	Accessories	2023-07-02
O003	Mary Jones	London	Keyboard	Accessories	2023-07-03
O004	Mary Jones	London	Laptop	Electronics	2023-07-04

Issues: Redundant data (e.g., repeated CustomerName and CustomerCity), which normalization will address.

Normalized Tables (Post-Normalization)

To demonstrate normalization, the `RawSalesData` table is split into a star schema with fact and dimension tables.

FactSales Table

OrderID	CustomerID	ProductID	SaleDate	Quantity	Price
O001	C1	P1	2023-07-01	2	1200.00
O002	C1	P2	2023-07-02	5	25.00
O003	C2	P3	2023-07-03	3	45.00
O004	C2	P1	2023-07-04	1	1200.00

DimCustomer Table

CustomerID	CustomerName	CustomerCity
C1	John Smith	New York

CustomerID	CustomerName	CustomerCity
C2	Mary Jones	London

DimProduct Table

ProductID	ProductName	ProductCategory
P1	Laptop	Electronics
P2	Mouse	Accessories
P3	Keyboard	Accessories

Instructions to Use the Dataset

1. Save the `FactSales` , `DimCustomer` , and `DimProduct` tables as CSV files (e.g., `FactSales.csv` , `DimCustomer.csv` , `DimProduct.csv`).
2. Import them into Power BI using "Get Data" > "Text/CSV" for the exercises below.

Explanations

What is Data Modeling?

Data modeling in Power BI involves structuring data tables to optimize performance, reduce redundancy, and enable efficient analysis. It defines how tables are related and organized to support interactive visualizations and calculations. A well-designed data model, such as a star schema, ensures scalability and simplifies queries for datasets like AdventureWorks 2022.

Normalization

Normalization organizes data into multiple related tables to eliminate redundancy and maintain data integrity. It follows rules like:

- **First Normal Form (1NF):** Ensure no repeating groups; each column contains atomic values.
- **Second Normal Form (2NF):** Remove partial dependencies; non-key attributes depend

on the entire primary key.

- **Third Normal Form (3NF):** Eliminate transitive dependencies; non-key attributes depend only on the primary key.

For example, splitting `RawSalesData` into `FactSales`, `DimCustomer`, and `DimProduct` reduces redundancy (e.g., repeated customer names).

Star Schema Design

A star schema consists of:

- **Fact Table:** Contains measurable data (e.g., sales quantities, prices) with foreign keys linking to dimension tables.
- **Dimension Tables:** Store descriptive data (e.g., customer or product details) with primary keys.

In AdventureWorks 2022, `Sales.SalesOrderDetail` (fact table) connects to `Production.Product` and `Sales.SalesOrderHeader` (dimension tables) via keys like `ProductID` and `SalesOrderID`.

Building Relationships

Relationships link tables using primary and foreign keys in Power BI's Model View. Types include:

- **One-to-Many:** Common in star schemas (e.g., one `ProductID` in `Production.Product` links to many rows in `Sales.SalesOrderDetail`).
- **Many-to-One:** The reverse of one-to-many (e.g., many sales orders link to one customer).
- **One-to-One:** Rare, used for unique mappings (e.g., one employee to one personnel record).

Relationships enable integrated analysis across tables, such as calculating total sales by product category.

Examples

Example 1: Normalizing RawSalesData

1. Import `RawSalesData.csv` into Power Query Editor.

2. Create `DimCustomer` by selecting `CustomerName` and `CustomerCity`, removing duplicates, and adding `CustomerID` (e.g., C1, C2).
3. Create `DimProduct` by selecting `ProductName` and `ProductCategory`, removing duplicates, and adding `ProductID` (e.g., P1, P2).
4. Create `FactSales` by merging with `DimCustomer` and `DimProduct` to replace names with IDs, keeping `OrderID`, `CustomerID`, `ProductID`, `SaleDate`, `Quantity`, and `Price`.
5. Load the normalized tables into Power BI and create relationships.

Example 2: Building a Star Schema with AdventureWorks 2022

1. Import `Sales.SalesOrderDetail` (fact table), `Production.Product`, and `Sales.SalesOrderHeader` (dimension tables) from AdventureWorks 2022.
2. In Model View, create a one-to-many relationship from `Production.Product` (`ProductID`) to `Sales.SalesOrderDetail` (`ProductID`).
3. Create a one-to-many relationship from `Sales.SalesOrderHeader` (`SalesOrderID`) to `Sales.SalesOrderDetail` (`SalesOrderID`).
4. Verify relationships in Model View to ensure a star schema.

Example 3: Creating Relationships with Custom Dataset

1. Import `FactSales.csv`, `DimCustomer.csv`, and `DimProduct.csv` into Power BI.
2. In Model View, create a one-to-many relationship from `DimCustomer` (`CustomerID`) to `FactSales` (`CustomerID`).
3. Create a one-to-many relationship from `DimProduct` (`ProductID`) to `FactSales` (`ProductID`).
4. Create a table visual showing `CustomerName`, `ProductName`, and `Quantity` to test the relationships.

Classwork

1. Task 1: Normalize RawSalesData

- Import `RawSalesData.csv` into Power Query Editor.
- Create `DimCustomer` and `DimProduct` tables by removing duplicates and adding IDs.
- Create `FactSales` with foreign keys and load all tables into Power BI.

2. Task 2: Build Star Schema with AdventureWorks

- Import Sales.SalesOrderDetail, Production.Product, and Person.Person from AdventureWorks 2022.
 - Create one-to-many relationships using ProductID and BusinessEntityID.
 - Create a table visual showing ProductName and LineTotal.
3. **Task 3: Create Relationships with Custom Dataset**
- Import FactSales.csv, DimCustomer.csv, and DimProduct.csv.
 - In Model View, create relationships between FactSales and the dimension tables.
 - Create a bar chart showing total Price by ProductCategory.
4. **Task 4: Data Profiling for Relationships**
- Import Sales.SalesOrderHeader from AdventureWorks 2022.
 - Use Column Profile to check the uniqueness of SalesOrderID and CustomerID.
 - Document findings to confirm suitability for relationships.

Scenarios

1. **Scenario 1: Retail Data Modeling**
- A retail company uses RawSalesData.csv with redundant data. Normalize the dataset into fact and dimension tables, create relationships in Model View, and build a table visual showing CustomerName, ProductName, and Price. Document the normalization steps and verify relationships.
2. **Scenario 2: AdventureWorks Sales Analysis**
- A manager needs to analyze sales by product and customer using AdventureWorks 2022. Import Sales.SalesOrderDetail, Production.Product, and Sales.Customer. Build a star schema with relationships on ProductID and CustomerID. Create a bar chart showing total LineTotal by ProductName and describe the data model.
3. **Scenario 3: Custom Dataset Analysis**
- The company wants to analyze sales by city using FactSales.csv, DimCustomer.csv, and DimProduct.csv. Create relationships, then build a table visual showing CustomerCity, ProductCategory, and total Quantity. Document the steps and insights.

Class Test

1. What is the primary goal of data modeling in Power BI?
- A) Creating visualizations

- B) Optimizing data structure for analysis
- C) Importing data from sources
- D) Sharing reports online

Answer: B

2. Which table in a star schema contains quantitative data?

- A) Dimension Table
- B) Fact Table
- C) Lookup Table
- D) Bridge Table

Answer: B

3. What does normalization aim to reduce?

- A) Data accuracy
- B) Data redundancy
- C) Data relationships
- D) Data types

Answer: B

4. Which relationship type is most common in a star schema?

- A) One-to-One
- B) One-to-Many
- C) Many-to-Many
- D) No Relationship

Answer: B

5. **Practical Task:**

- Import `FactSales.csv` , `DimCustomer.csv` , and `DimProduct.csv` into Power BI.
- Create one-to-many relationships from DimCustomer (CustomerID) and DimProduct (ProductID) to FactSales.
- Create a table visual showing CustomerName, ProductName, and Quantity.
- Submit a screenshot of the visual and list the steps taken to create the relationships.